



TUDOR ANDREI HĂLĂȘAG

ROBOTICS & AUTOMATION ENGINEER

Details

Brasov
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Links

tudor-halasag.github.io
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Skills

C / C++
Python
MATLAB
Arduino
AutoCAD Mechanical
Vericut
LabVIEW
SAP
KUKA Robotics
CNC Machining
Root Cause Analysis
Microsoft Office Suite

Languages

Romanian — Native
English — C2
Italian — A2
German — A1
Turkish — A1

Interests

Motorsport filming & carspotting
Music — drums, guitar, bass, DJ
Capital markets & investing
Hiking & outdoor fitness

Professional Summary

I take things apart to understand them — then engineer them better. Robotics and Industrial Engineering student with hands-on experience in CNC machining, KUKA robotics, automation, and embedded systems, gained through direct work in an industrial manufacturing environment. I learn every process from the ground up before I try to improve it — that has led to designing custom tooling, specifying robotic work cells, and driving automation initiatives from data collection to supplier evaluation. Looking to build a career engineering the systems behind real, physical production.

PROJECTS & PERSONAL DEVELOPMENT

AGV Swarf Transport Feasibility Study, Airbus Aerostructures

Led field data collection across 58 CNC machines to map manual swarf-cart transport workflows; defined process parameters and authored a structured technical RFI to evaluate AGV automation solutions with suppliers.

Vice-Tightening Tool Design , Airbus Aerostructures

Designed a custom tool and bit set with impact-gun attachment to replace manual wrench tightening on CNC vices, targeting precise 75 N-F clamping force; tested prototypes and validated in pilot phase.

StepDown Transformer Design

Designed a functional stepdown transformer from scratch, applying electromagnetic theory and component selection.

Arduino & Embedded Systems Lab

Programmed microcontrollers and interfaced LCD displays, sensors, stepper motors, and speakers — handson embedded I/O and memory management.

EDUCATION

BSc - Robotics at the Faculty of Electrical Engineering and Computer Science, Transilvania University (UNITBV), Brasov

2026

- Transferred from UPB to continue studies closer to an industry placement in Braov.
- Control systems, robotics, and industrial automation curriculum.

BSc - Faculty of Robotics and Industrial Engineering, Polytechnic University of Bucharest, Bucharest

2024 — 2026

- Control systems, robotics, and industrial automation — with handson lab projects.
- Systemthinking applied to real engineering problems across hardware and software disciplines.

Mathematics - Computer Science, "MIHAIL KOGLNICEANU" NATIONAL COLLEGE, Galati

2017 — 2021

Final Exam: 8.76 / 10

- Intensive track in advanced mathematics and programming.
- Practical C++ and Python from the earliest stages — applied in personal projects throughout.

EMPLOYMENT HISTORY

Production Technologist Intern, Airbus Aerostructures, Ghimbav

JUL 2026

- Co-specified an automated KUKA robot cell for a 4-machine CNC cluster, defining requirements for heavy-part handling, automated 180° part rotation, and an integrated magnetic tool-change system to minimize operator intervention.
- Owned quality assurance across CNC (Grob 350/550) and robotic (KUKA) cells — first-shift inspections, fixture/setup verification, SAP non-conformance tracking, and root-cause troubleshooting of robotic motion errors that prevented large-batch scrap.
- Redesigned the deburring zone workflow (FIFO layout, ergonomic electric scissor-lift carts, roller shelving) and co-developed the technical spec for an automated deburring robot cell, pending management approval.
- Rewrote the official KUKA cell work instruction after mastering the process hands-on — guided operator work at milling/deburring stations and vice assembly under the shop foreman.

Barista & Operations Lead, 5 TO GO EUROPE HOLDING SRL, Romania, Bucharest

MAR 2023 — SEPT 2023

- Managed all transactions and inventory via Comax Smart Retail (ERP/POS) and Excel — 100% cashhandling accuracy throughout.
- Drove a 15–20% improvement in peakhour service efficiency through process restructuring and preparation systems.